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**A Logistic Regression Model:
Predicting Coffee Quality Tier without Cupping**

Course: Data Analysis 1
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I. Introduction

Coffee is one of the most widely consumed beverages and agricultural exports for many developing countries. A coffee’s quality grade (Commercial, Premium, or Specialty) has significant economic consequences for farmers, exporters, importers, and roasters. Being able to predict coffee quality early in the supply chain can help stakeholders make better sourcing decisions, reduce waste, and reward high-quality producers with appropriate pricing.

This project aims to develop and evaluate multiclass classification models that predict the quality tier of Arabica coffee samples using two types of features:

1. Production-level attributes such as growing altitude, processing method, bean variety, and physical defects.
2. Sensory evaluation scores assigned by certified Q-graders during cupping sessions.

By comparing these two feature sets, we aim to determine how well coffee quality tier can be predicted from production-side features alone, without any cupping data, and to explore the implications for the real-world applications such as pre-screening, pricing, and farmer self-assessment.

The dataset used in the project is the Coffee Quality Institute (CQI) Arabica Quality Database (LeDoux, 2018), a public dataset that contains 1,311 professionally evaluated Arabica coffee samples from 34 countries, spanning harvest years from 2009 to 2018. Each record includes origin information, processing details, physical defect counts, altitude data, and 10 individual sensory scores (Aroma, Flavor, Aftertaste, Acidity, Body, Balance, Uniformity, Clean Cup, Sweetness, and Cupper Points), each scored on a 1–10 scale by certified Q-graders, which together sum to the Total Cup Points score (maximum 100). From Total Cup Points, we derived our target group `quality_tier` using the Specialty Coffee Association’s standard grading threshold: Commercial (below 80), Premium (80–84), and Specialty (85 and above) (Specialty Coffee Association, 2005).

After preprocessing, which primarily includes parsing messy altitude and harvest year fields, handling missing values, and dropping a small number of rows with unparseable data, this dataset contains 1,251 Arabica coffee samples across three different quality tiers: Commercial (14%), Premium (78%), and Specialty (8%).

II. Data Preprocessing

The raw CQI dataset contains 1,311 rows and 44 columns mixing sensory scores, production metadata, administrative identifiers, and pre-computed cleanup fields of uneven quality. This section documents every step of our cleaning process in the order it was performed in the notebook.

II.0. Initial Data Inspection

The dataset was loaded from the CQI Github Repository using `pd.read_csv()`, returning 1,311 rows and 44 columns. From there, we converted all the column names to `snake_case` format to ensure consistency throughout our whole analysis. We then computed an initial column-level summary using `df.dtypes`, `df.nunique()`, and `df.isnull().sum()` to capture data types, cardinality, and missingness for every column.

Calling `df.duplicated().sum()` confirmed that there were no duplicate rows. However, many columns had missing values, which in itself also had a clear pattern: the columns with the highest proportion of missing values were administrative identifiers; `lot_number` (79.4% missing), `farm_name` (27.2%), `mill` (23.7%), `producer` (17.5%), and `company` (15.9%). This pattern is consistent with the CQI submission process, where producers may submit coffees without complete administrative paperwork.

Among coffee-attribute columns, missingness appeared in `color` (20.4%), the three altitude-related fields (each 17.3% missing), the raw `altitude` column (17.0%), `variety` (15.3%), and `processing_method` (11.6%). Each of these required individual handling decisions in subsequent preprocessing steps.

Most data types were appropriate (the ten sensory scores stored as `float64`, defects count as `int64`), but two columns stored as `object` required custom parsing: `altitude` because of mixed unit formats (e.g., “1500 m”, “4200 feet”, “1000-1500m”), and `harvest_year` because of inconsistent formatting (year ranges, quarter notation, multilingual month names). The pre-computed altitude cleanup columns (`altitude_low_meters`, `altitude_high_meters`, and `altitude_mean_meters`) appeared clean as `float64` but, as we discovered later in [Section II.3](#), contained systematic parsing bugs.

II.1. Drop Irrelevant Columns

Based on the inspection in the previous section, we identified 23 columns that were not useful for predicting coffee quality and removed them before further analysis. These fell into four categories:

Zero-variance column. The `species` column contained only one unique value “Arabica” for all the rows. Since the dataset itself is restricted to Arabica coffee, this column is useless and was dropped.

Administrative identifiers. Thirteen columns recorded paperwork metadata: `unnamed:_0` (a stale row-index column from the original CSV), `lot_number`, `ico_number`, `farm_name`, `mill`, `producer`, `company`, `owner`, `owner_1`, `in_country_partner`, `certification_body`, `certification_address`, and `certification_contact`. These columns identified who produced or certified each coffee, not attributes of the coffee itself. Most had hundreds of unique values (for example, `farm_name` had 557 unique values, `mill` had 447, and `lot_number` had 221 with 79.4% missingness). Including these columns in our model would force it to memorize specific producers or certification bodies rather than learning generalizable patterns about coffee quality. Because our goal is to predict tier from features that generalizes to coffees from any producer and origin, these columns were excluded.

Process metadata. Four columns recorded certification logistics or batch information rather than coffee properties: `grading_date` and `expiration` describe when each sample was processed, while `bag_weight` and `number_of_bags` describe the size of the submission batch.

Redundant or buggy columns. Five columns either duplicated information we would extract ourselves or contained errors. The four altitude-related columns (`unit_of_measurement`, `altitude_low_meters`, `altitude_high_meters`, and `altitude_mean_meters`) were the attempted cleanup of the raw `altitude` text but contained systematic parsing bugs, as discussed in [Section II.3](#). The `region` column had 343 unique country region names with most having only 1–2 samples, meaning it had too high cardinality for the model. Instead, we capture geographic information through the `country_of_origin` column which we later regrouped into broader continental regions in [Section II.8](#).

After this drop, the working dataset contained **1,311 rows** and **21 columns**, ready for targeted processing in the rest of [Section II](#).

II.2. Initial Processing Decisions

After dropping irrelevant columns, we re-examined missingness across the remaining 21 columns to confirm which ones require targeted cleaning. The missing pattern persists: `color` (267 missing, 20.4%), the raw `altitude` column (223 missing, 17.0%), `variety` (201 missing, 15.3%), `processing_method` (152 missing, 11.6%), and `harvest_year` (47 missing, 3.6%) with single missing values in `country_of_origin` and `quakers`. Each of these is handled in its own subsection below ([II.3](#) through [II.10](#)).

We also identified one row with `total_cup_points == 0`, which is impossibly low. The practical minimum cupping score is around 60, and so we treated this row as a data-entry error and dropped it using `df = df[df['total_cup_points'] > 0]`, leaving us with 1,310 rows.

Finally, we used `total_cup_points` to construct the classification target `quality_tier` using the Specialty Coffee Association’s standard grading thresholds:

- Commercial: total cup points below 80
- Premium: total cup points between 80 and 84.99
- Specialty: total cup points 85 or above

Applied to our data, the target distribution was 174 Commercial samples (13.3%), 1,033 Premium samples (78.9%), and 103 Specialty samples (7.9%). This huge class imbalance shaped several of our decisions in model development in [Section IV](#).

II.3. Altitude Parser

The `altitude` column required the most preprocessing effort. The dataset provided four altitude cleanup columns (`unit_of_measurement`, `altitude_low_meters`, `altitude_high_meters`, and `altitude_mean_meters`), but these contained systematic bugs that made them unreliable. Therefore, we had to create a customized parser using the raw `altitude` text column.

Inspections of `altitude_mean_meters` revealed impossible values: the maximum was **190,164 meters** (for reference, Mount Everest’s summit is **8,849 meters tall**). This is also far above the plausible coffee growing range of **100–3,000 meters**. Filtering rows where `altitude_mean_meters > 3000` returned 14 rows that fell into two problem categories:

- **Decimal-stripping errors:** Two rows showed definite parsing bugs: row 1144 had a raw altitude value of “**1901.64**” (Guatemala) but was stored as **190164.0** in the cleanup columns, and row 1040 had a raw value of “**1100.00 mosl**” (mosl here is likely a typo of masl, which stands for meters above sea level) (Nicaragua) stored as **110000.0**. In both cases, the parser had stripped the decimal point and treated the digits as a single integer, wrongly multiplying the altitude by 100.
- **Producer-side reporting errors:** The remaining 12 implausible rows reflect submission errors by producers themselves rather than parser bugs. Several Guatemalan rows reported “3280” with `unit_of_measurement = 'm'`, recording a producer’s likely confusion between feet (~1000m equivalent) and meters. Multiple Myanmar rows reported altitudes between 3,800 and 4,287 m, which are biologically impossible, and one Brazilian row reported “11000 metros”.

We wrote a `parse_altitude()` function that processed each raw altitude string in four steps:

1. Extract all numeric values using a regular expression `\d+\.\d*`, which captures both integers and decimals (preventing the decimal-stripping bug).
2. If two numbers are extracted (e.g., “1000-1500m”), use the midpoint.
3. If the string contains `feet`, `ft`, or an apostrophe, convert the value from feet to meters using the factor 0.3048.
4. If the resulting altitude falls outside the biologically plausible range of 100–3000 meters, flag the value as missing.

After parsing, 948 rows had valid altitudes; 360 rows had either missing original values or had been flagged by step 4 of our parser. Rather than dropping these rows or imputing with a global mean (which would mess up regional patterns), we imputed each missing altitude with the median altitude of valid samples from the same `country_of_origin`. This preserved real geographic signal rather than collapsing all missing values to a single global average, however two rows could not be imputed and were dropped as their `country_of_origin` was missing, leaving **1,308 rows**.

The final cleaned `altitude_imputed_m` column has a mean of 1,301 m, median of 1,350 m, and standard deviation of 398 m, which makes sense realistically for global Arabica production.

II.4. Harvest Year Parser

The `harvest_year` column was stored as `object` rather than numeric type because of inconsistent submission formats. The 46 unique values fell into five formatting patterns:

- **Plain years:** Standard 4-digit years like “2012” or “2015”.
- **Year ranges:** Coffee harvest can span calendar years, so producers submitted ranges such as “2013/2014” or “2009 - 2010”.
- **Two-digit year codes:** A few rows used abbreviated formats like “08/09 crop” or “4T/10”.
- **Quarter notation:** Trimester format (“1T/2011” or “4T/2010”, where “T” stands for Trimester).
- **Date text in multiple languages:** English (“January 2011” or “23 July 2010”), Spanish (“Abril - Julio /2011”). Some entries lacked any year reference at all: “Abril - Julio”, “Mayo a Julio”, “May-August”, “August to December”, “January Through April”, and two outright junk entries “TEST” and “mmm”.

We wrote a `parse_harvest_year()` function with two regex-based strategies:

1. **Find any 4-digit year in the plausible range 2000–2029** using the regex `\b(20[0-2]\d)\b`. If multiple years are found, take the later year.
2. **Fallback for two-digit codes:** If no 4-digit year is found, search for 2-digit codes following a slash using `/(\d{2})\b` (e.g., “4T/10” → 10 → 2010).
3. **Junk handling:** Strings identified as having no extractable year information return `NaN`.

The parser successfully extracted years for 1,251 of 1,308 rows. The 57 unparseable rows were dropped for subsequent analysis. The cleaned `harvest_year_clean` column spans 2009 to 2018 with a median of 2014 and standard deviation of 1.85 years.

II.5. Variety Grouping

The `variety` column contained 29 unique values plus 156 missing entries (12.5% missingness). To reduce cardinality while preserving meaningful information, we decided to group varieties into 10 botanical families using the Arabica variety taxonomy maintained by World Coffee Research and the Specialty Coffee Association.

The grouping logic was as follow:

- **Bourbon:** Bourbon and Yellow Bourbon.
- **Typica:** Typica, Mundo Novo, Blue Mountain, and Java.
- **SL_Kenyan:** SL14, SL28, and SL34.
- **Specialty:** Geisha, Pacamara, Pacas, Marigojipe, Peaberry, and Moka Peaberry (rare specialty varieties, each with fewer than 15 samples in our dataset).
- **Caturra, Catuai, Catimor, and Hawaiian Kona:** Kept as their own groups, as each had at least 20 samples and a distinct cultivation identity.
- **Other:** Remaining low-sample varieties (Arusha, Ruiru 11, Sumatra, Mandheling, Sumatra Lintong, Ethiopian Yirgacheffe, Ethiopian Heirlooms, Sulawesi, Pache Comun, and entries originally labelled as “Other”). These are mostly specialty or regional varieties with only 1–5 samples each.
- **Unknown:** The 156 missing entries, preserved as their own informative category. We treated this missingness as a feature rather than imputing it, since absent variety information may itself correlate with submission characteristics that distinguish quality tiers.

The final distribution across the 10 groups was: Bourbon (259 samples), Caturra (252), Typica (247), Unknown (156), Other (127), Catuai (72), Hawaiian Kona (44), SL_Kenyan (38), Specialty (36), and Catimor (20). Each group had a sample size of at least 20, supporting reliable estimation in the chi-square test (Section III.8) and the logistic regression model (Section IV).

II.6. Processing Method Cleanup

The `processing_method` column had five unique values plus 100 missing entries (8.0% missingness):

- **Washed / Wet** – 808 samples
- **Natural / Dry** – 249 samples
- **Semi-washed / Semi-pulped** – 56 samples
- **Pulped natural / honey** – 12 samples
- **Other** – 26 samples

Two issues required handling. First we shortened the labels containing the slash notation for readability (i.e. **Washed / Wet** → **Washed**, **Natural / Dry** → **Natural**, and **Semi-washed / Semi-pulped** → **Semi-washed**). Second, the **Pulped natural / honey** category contained only 12 samples, so we merged them into the **Semi-washed** category, as their process involved partial pulp removal. The 100 missing entries were labelled **Unknown** rather than imputed, applying the same logic as `variety_group` in [Section II.5](#).

The final distribution was: **Washed** (808 samples, 64.6%), **Natural** (249, 19.9%), **Unknown** (100, 8.0%), **Semi-washed** (68, 5.4%) and **Other** (26, 2.1%). All five categories had more than 25 samples.

II.7. Color Cleanup

The `color` column contained three unique values plus 220 missing entries (17.6% missingness):

- **Green** – 839 samples, 67.07%
- **Bluish-Green** – 112 samples, 8.95%
- **Blue-Green** – 80 samples, 6.39%

The distinction between **Bluish-Green** and **Blue-Green** is linguistic rather than meaningful; both of them describe freshly-milled, high-moisture green coffee beans whose color reflects recent processing. We merged them into a single **Blue-Green** category since they represent the same physical state of the coffee beans.

The 220 missing entries were labelled **Unknown**, applying the same convention as `variety_group` and `processing_group` in earlier sections.

The final three-category distribution was: **Green** (839 samples, 67.1%), **Unknown** (220 samples, 17.6%), and **Blue-Green** (192 samples, 15.4%).

II.8. Country to Region Grouping

The `country_of_origin` column contained 34 unique values with severe imbalance: the top five countries accounted for 62% of samples, while 15 countries had fewer than 10 samples each. We therefore grouped countries into five coffee-growing regions based on the International Coffee Organization’s geographic classifications for coffee-exporting nations.

The grouping logic was:

- **Central America:** Mexico, Guatemala, Honduras, Costa Rica, Nicaragua, El Salvador, Panama. These countries share volcanic highland terrain, predominantly **Washed** processing, and similar **Bourbon-Caturra-Catuai** variety mixes.
- **South America:** Colombia, Brazil, Peru, Ecuador. Large-scale production with mixed processing methods.
- **East Africa:** Ethiopia, Tanzania, Uganda, Kenya, Malawi, Burundi, Rwanda, Zambia, and Côte d’Ivoire. The Arabica birthplace, characterized by high altitudes and heirloom varieties. Côte d’Ivoire (one sample) was the only West African country in our dataset and was merged into East Africa as the closest analogous region.

- **Southeast Asia:** Taiwan, Thailand, Indonesia, China, Myanmar, Vietnam, Philippines, Laos, and Papua New Guinea. Predominantly Robusta producers globally, but the Arabica samples in this dataset reflect specialty cultivation in regions like Sumatra, Yunnan (China), and northern Thailand.
- **Caribbean/US:** Hawaii, Haiti, Puerto Rico, U.S. mainland, and Mauritius. Small-volume origins with distinctive terroir; Hawaiian Kona dominates this group with 54 samples.

The final five-region distribution was: **Central America** (561 samples, 44.8%), **South America** (309, 24.7%), **Southeast Asia** (161, 12.9%), **East Africa** (147, 11.8%), and **Caribbean/US** (73, 5.8%). All regions exceeded 70 samples, supporting reliable estimation in the chi-square test ([Section III.8](#)) and the logistic regression model ([Section IV](#)).

II.9. Moisture Cleanup

The `moisture` column appeared complete at first glance, as it had no missing values. However, descriptive statistics revealed an unusual pattern: 229 rows (18.3% of the data) had a moisture reading of exactly 0.

Zero moisture is physically impossible for coffee beans. They are processed to a target of around 10 to 12 per cent for shelf stability, and the SCA’s green grading protocol specifies that moisture below 9% indicates over-drying, while moisture above 13% risks mould growth. We therefore interpreted these 229 zero values as hidden missingness rather than legitimate readings.

We chose to impute these values rather than drop the rows. Two imputation strategies were considered:

1. **Country-level median** (used in [Section II.3](#) for altitude). This would preserve regional processing differences if they exist.
2. **Global median.** Simpler, but treats all coffees identically regardless of region of origin.

To choose between the two methods, we computed the median moisture by region using only non-zero values and got the following results:

Region	Median Moisture
Caribbean/US	0.110
Central America	0.110
East Africa	0.120
South America	0.110
Southeast Asia	0.110

Because regional medians for moisture had a spread of just 0.01 (1 percentage point), we decided to use the simpler global median (0.110 or 11.0%) to fill all 229 zero values.

After imputation, the `moisture` column had a minimum of 0.01, median of 0.11, maximum of 0.28, and standard deviation of 0.018, which is consistent with industry norms.

II.10. Quakers Missing Value

The `quakers` column contained a single missing value (out of 1,251 rows), which we handled by using the mode 0. This is because of two reasons:

1. **Modal imputation matches the dominant pattern.** With over 92% of samples at zero, imputing the single missing value with zero is a low-risk choice.
2. **The single-row impact is negligible.** Whatever value we imputed, its effect of summary statistics, bivariate analysis, and model development would be practically invisible at our sample size of 1,251.

After this fill, the `quakers` column had no missing values, with summary statistics: minimum 0, mean 0.17, maximum 11, and standard deviation 0.8.

II.11. Drop color_group

After cleaning the four categorical features (`variety_group`, `processing_group`, `color_group`, `region_group`), we ran chi-square tests of independence between each feature and `quality_tier` to assess relative association strength. The full chi-square results are reported in [Section III.8](#) as part of the bivariate analysis, but we use the ranking here to justify a final preprocessing decision.

The chi-square statistic ranked the four features as follows: `region_group` (188.98) > `variety_group` (126.41) > `processing_group` (47.78) > `color_group` (42.08). All were statistically significant at $p < 0.001$, but with $n = 1,251$, statistical significance is near-automatic — the chi-square statistic itself is the more meaningful comparison.

`color_group` showed the weakest association with quality tier, the highest missingness of any retained categorical feature (17.6%), and only three categories (Green, Blue-Green, Unknown) of which one is the missing-data placeholder. Combined with the fact that bean color is largely a function of recent processing rather than an independent quality signal, we dropped `color_group` from the feature set. After this drop, the working dataset contained 1,251 rows and 20 columns, ready for exploratory analysis in [Section III](#).

III. Exploratory Data Analysis

III.1. Target Distribution

The dataset is highly imbalanced, with the following distribution:

- Premium Class: 78%
- Specialty: 8.1%
- Commercial: 13.9%

To address this imbalance, `class_weight='balanced'` was applied in one of the logistic regression models (Model A – balance) to ensure better performance across all classes.

III.2. Numeric Production Features

Feature	Count	Mean	SD	Min	25%	Median	Max
<code>altitude_imputed_m</code>	1,251	1,310.7	390.0	100.0	1,054.5	1,350.0	2,560.0
<code>harvest_year_clean</code>	1,251	2,013.7	1.8	2,009	2,012	2,014	2,018
<code>moisture</code>	1,251	0.111	0.018	0.01	0.11	0.11	0.28
<code>category_one_defects</code>	1,251	0.42	1.84	0	0	0	31
<code>category_two_defects</code>	1,251	3.66	5.41	0	0	2	55
<code>quakers</code>	1,251	0.17	0.80	0	0	0	11

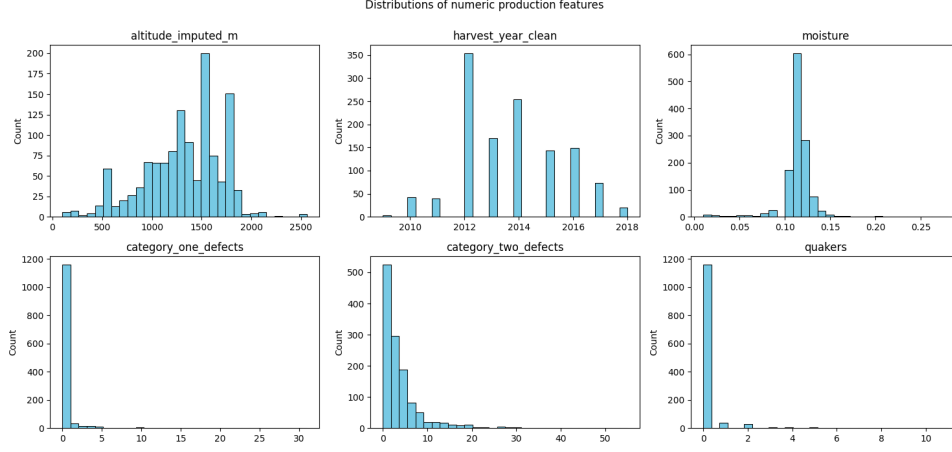


Figure 1: Distribution of Numeric Production Features

- **Altitude (`altitude_imputed_m`)**: The average altitude is 1,311 m, with most samples between 1,054 m and 1,600 m. Some reach up to 2,560 m. The data is slightly skewed toward higher altitudes, which is normal for Arabica coffee growing areas.
- **Harvest year (`harvest_year_clean`)**: Coffee samples range from 2009 to 2018, with most collected between 2012 and 2015. The average year is 2013.7, showing the dataset mainly covers that period.
- **Moisture**: The average moisture level is about 11.1%, and most values fall between 11% and 12%, which matches standard industry levels. A few very low values (around 1%) are likely unusual outliers.
- **Category one defects**: Most coffees have no primary defects at all (median = 0). However, a small number have very high defect counts (up to 31), making the distribution heavily skewed.
- **Category two defects**: Most samples have low defect counts, but some have much higher values (up to 55). The average is 3.66, but many coffees are still close to zero, with a few extreme cases increasing the spread.
- **Quakers**: Most samples have zero quakers (median = 0, mean = 0.17), with a few high values. Due to low variation and weak correlation with the target, `quakers` and `harvest_year_clean` were excluded from the final model.

III.3. Sensory Scores

Feature	Count	Mean	SD	Min	25%	Median	Max
aroma	1,251	7.57	0.32	5.08	7.42	7.58	8.75
flavor	1,251	7.52	0.34	6.08	7.33	7.50	8.83
aftertaste	1,251	7.40	0.35	6.17	7.25	7.42	8.67
acidity	1,251	7.54	0.32	5.25	7.33	7.50	8.75
body	1,251	7.52	0.29	5.25	7.33	7.50	8.58
balance	1,251	7.52	0.35	6.08	7.33	7.50	8.75
uniformity	1,251	9.85	0.49	6.00	10.00	10.00	10.00
clean_cup	1,251	9.84	0.74	0.00	10.00	10.00	10.00
sweetness	1,251	9.91	0.45	1.33	10.00	10.00	10.00
cupper_points	1,251	7.50	0.43	5.17	7.25	7.50	8.67

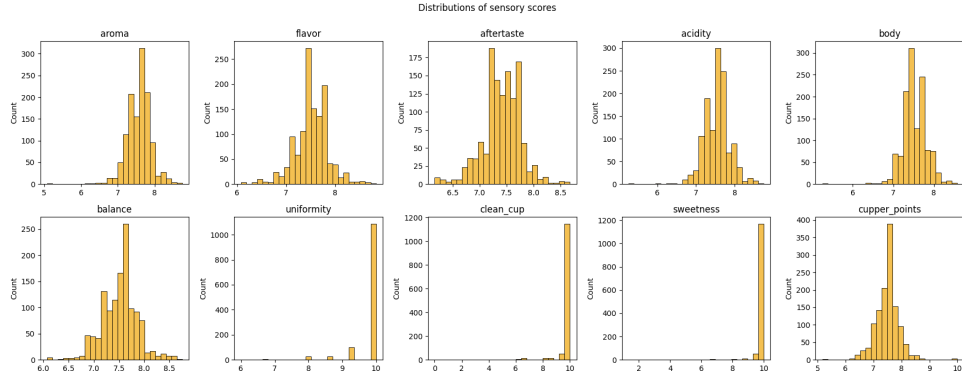


Figure 2: Distribution of Numeric Sensory Features

The ten sensory scores can be divided into two groups:

- Analytical scores (**aroma**, **flavor**, **aftertaste**, **acidity**, **body**, **balance**, **cupper_points**) mostly fall between 7.0 and 8.5, with standard deviations ranging from 0.29 to 0.43. These scores vary across samples, making them useful for identifying differences in coffee quality levels.
- Protocol scores (**uniformity**, **clean_cup**, **sweetness**) are generally very high, often close to 10, with mean values above 9.85 and standard deviations between 0.45 and 0.74. Since most samples receive similarly high scores, these features are less useful for distinguishing between different quality levels.

III.4. Categorical Features

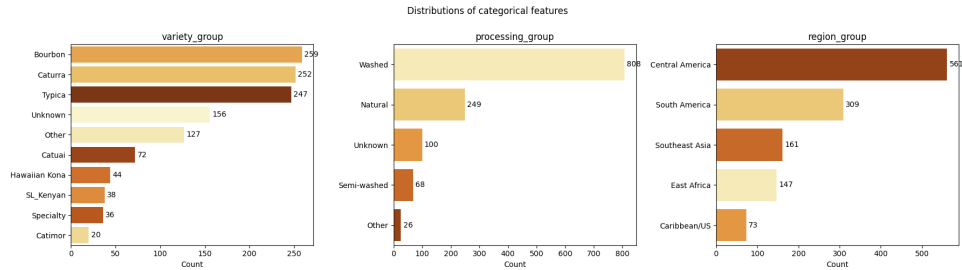


Figure 3: Distribution of Categorical Features

- Variety (**variety_group**): most samples come from Bourbon (259), Caturra (252), and Typica (247), making up around 60% of the dataset. Unknown (156) is the next largest group after these three groups, showing missing information in some records. Other varieties such as Other (127), Catuai (72), Hawaiian Kona (44), SL_Kenyan (38), Specialty (36) and Catimor (20) appear much less often. Since the dataset is mostly made up of three varieties, the model will learn these well, but it may struggle to learn patterns from the rare varieties.
- Processing method (**processing_group**): most coffee is processed using the Washed method (808). Natural processing is much less common (249), followed by Unknown (100), Semi-washed (68), and Other (26). This strong imbalance means the model will learn more about “Washed coffee”, but may not perform as well for less common processing methods.
- Region (**region_group**): Central America has the most samples (560), South America (309), followed by Southeast Asia (161), East Africa (147), and Caribbean/US (73). Even though East Africa has fewer samples, earlier analysis shows it produces the highest rate

of specialty coffee. Because of this imbalance, the model may not learn enough from East African data, even though it is very important for predicting high-quality coffee.

III.5. Numeric Features vs. Quality Tier

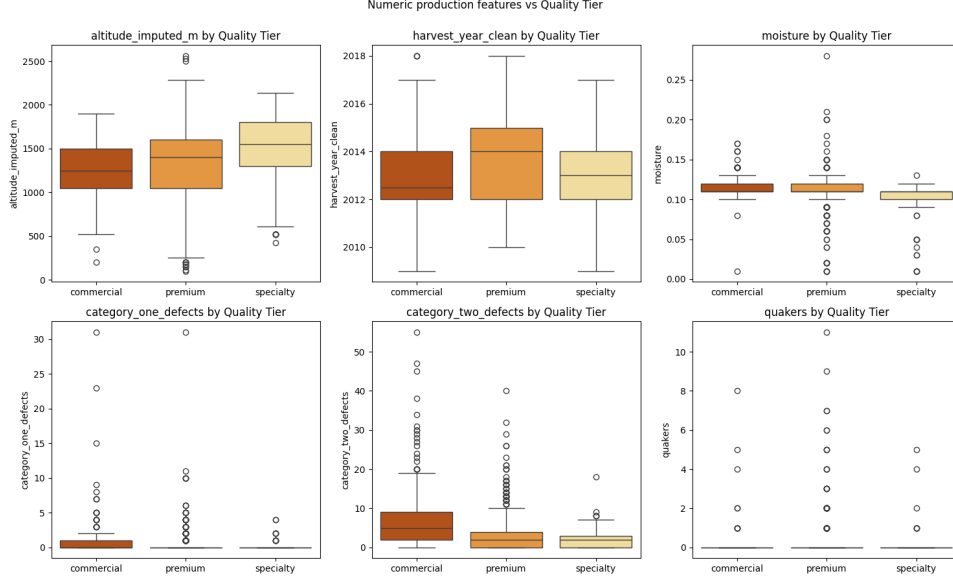


Figure 4: Numeric Features vs. Quality Tier

The boxplots show that:

- **Altitude (`altitude_imputed_m`):** Altitude increases as coffee quality improves. Specialty is grown at higher altitude and has a more consistent range compared to premium and commercial coffee. This suggests that higher altitude may improve coffee quality by slowing bean growth and improving flavor development (Wintgens, 2009).
- **Category one defects:** Commercial coffee has much higher defect counts than premium and specialty coffee. In contrast, Premium and Specialty coffees have very low and similar defect levels. This feature clearly separates lower-quality coffee from higher-quality coffee.
- **Category two defects:** Commercial coffee still shows higher defect levels, but the difference between groups is smaller than for category one defects. Premium coffee also shows more variation than specialty coffee.
- **Moisture:** All three quality groups have very similar moisture levels, with heavy overlap. This means moisture is not useful for distinguishing coffee quality.
- **Harvest year and quakers:** These features look almost the same across all quality groups, showing no clear differences. Therefore, they do not help in predicting quality.

III.6. Categorical Features vs. Quality Tier

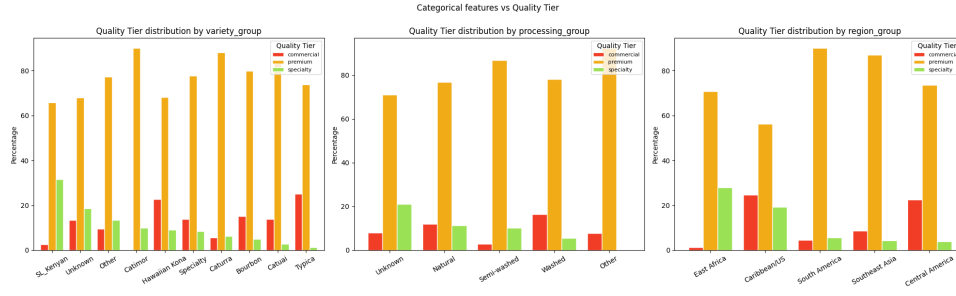


Figure 5: Categorical Features vs. Quality Tier

The grouped bar plots show how coffee quality tiers are distributed within each category:

- **Variety:** SL_Kenyan has the highest proportion of specialty coffee compared to other varieties. Catimor has the lowest specialty rate and more commercial coffee, which may be linked to its use as a commercial hybrid. Bourbon, Typica, and Caturra are mostly made up of premium coffee.
- **Processing method:** Natural-processed coffees tend to have more specialty coffee than Washed coffees, possibly because natural processing enhances flavor complexity. Semi-washed coffees are mostly premium.
- **Region:** East Africa has the highest proportion of specialty coffee among all regions. Caribbean/US also shows a relatively high specialty rate. Central America and South America are mostly premium coffee. Southeast Asia has the lowest specialty rate and a higher share of commercial coffee.

III.7. Correlation Heatmap

Pearson Correlation

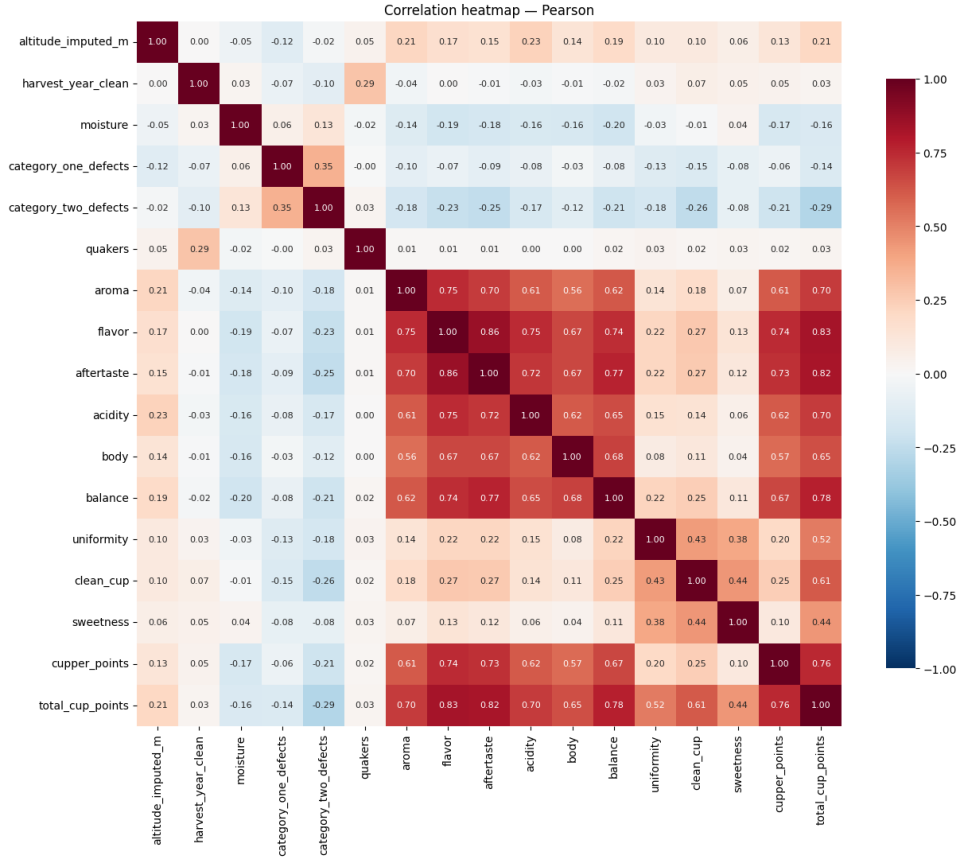


Figure 6: Correlation Heatmap – Pearson

- **Sensory scores are strongly related to each other:** The strongest relationships are flavor–aftertaste ($r = 0.86$), aftertaste–balance ($r = 0.77$), and aroma–flavor ($r = 0.75$). This shows that these scores reflect the same overall quality factor. Including all of them in one model could cause multicollinearity, which is why Model B uses only **flavor** as a representative feature.
- **Flavor is the strongest predictor of total_cup_points:** Flavor ($r = 0.83$), aftertaste ($r = 0.82$), and balance ($r = 0.78$) have the highest correlations with total_cup_points. In contrast, protocol scores (uniformity, clean_cup, sweetness) have weaker relationships ($r = 0.44$ – 0.61), so they were excluded from the model.
- **Production features show weak linear relationships:** The strongest production-related features are category_two_defects ($r = -0.29$) and altitude ($r = 0.21$), both in expected directions. Other features (moisture, category_one_defects, harvest_year_clean, quakers) have little to no correlation. This suggests that Model A, which uses only production features, is likely to have lower predictive performance than Model B.

Spearman Correlation

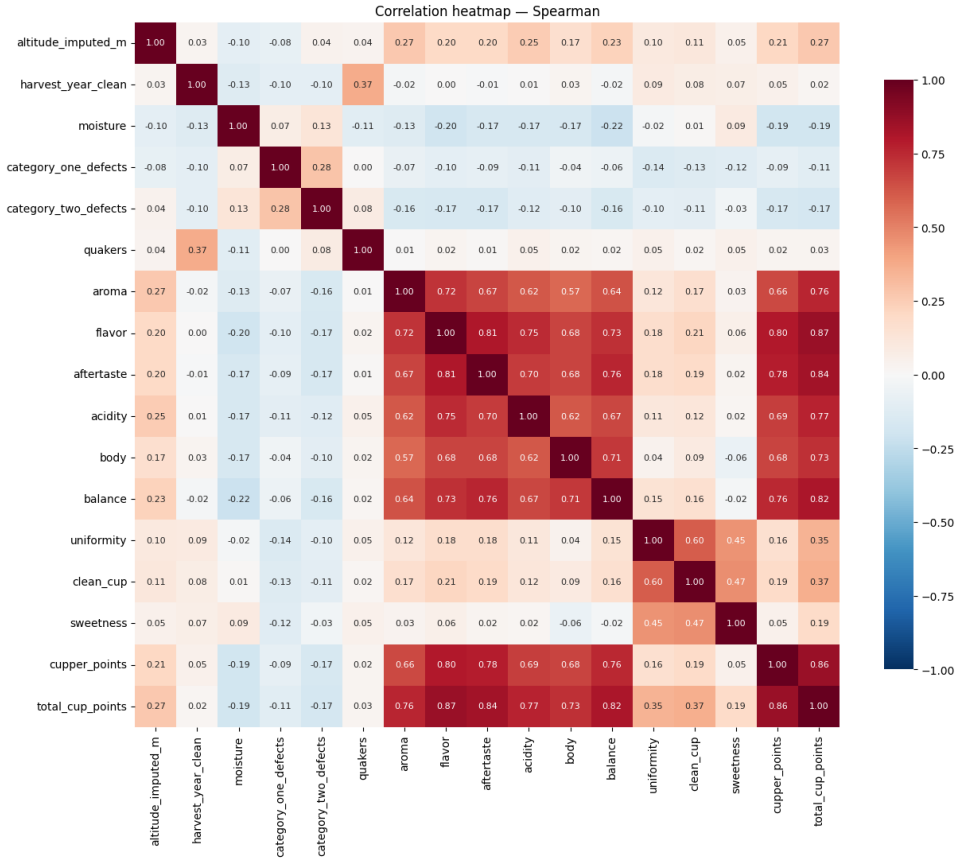


Figure 7: Correlation Heatmap – Spearman

- **Sensory relationships are consistent:** Spearman results are similar to Pearson, with strong relationships such as flavor–aftertaste ($r = 0.81$) and flavor–total_cup_points ($r = 0.87$). This confirms that these relationships are reliable and not dependent on linear assumptions.
- **Production features show mixed nonlinearity:** Spearman correlations for production features differ from Pearson in different ways (e.g., altitude rises from 0.21 to 0.27, suggesting a mild non-linear pattern, while category_two_defects drops in magnitude from -0.29 to -0.17 , suggesting Pearson was inflated by extreme defect outliers). Both methods agree on the qualitative ranking, with altitude positive and defects negative, confirming that the findings are robust to the choice of correlation method. This also indicates that logistic regression may not fully capture these non-linear effects, and more flexible models like Random Forest could perform better.
- **Harvest year and quakers remain irrelevant:** Both features show near-zero correlations in both methods, confirming they do not contribute useful information and should be excluded.

III.8. Chi-squared Tests

Chi-tests were used to check whether the four cleaned categorical features (variety_group, processing_group, region_group, and color_group) are related to the coffee quality tier. The chi-square statistic measures the strength of the association, while the p-value tells us whether the relationship is statistically significant.

Feature	Chi-square (χ^2)	df	p-value
region_group	188.98	8	1.34e−36 ***
variety_group	126.41	18	2.57e−18 ***
processing_group	47.78	8	1.09e−07 ***
color_group	42.08	4	1.60e−08 ***

(*** means $p < 0.001$, which is highly significant)

All four features showed statistically significant associations with quality tier at the $p < 0.001$ level, but with $n = 1,251$, even weak associations tend to come out as statistically significant. We therefore relied on the chi-square statistic itself as a relative ranking of association strength, recognizing that this comparison is approximate since chi-square values are also influenced by the number of categories.

As shown by the table above:

- **region_group** is the strongest predictor of quality tier, consistent with the visual pattern in Section III.6 where East Africa showed the highest specialty rate and Southeast Asia the lowest.
- **variety_group** also has a strong relationship with quality tier, matching the visual evidence that SL Kenyan varieties dominate the specialty tier while Catimor leans toward commercial.
- **processing_group** shows a weaker but still significant association, supporting the EDA observation that natural-processed coffees tilt slightly toward specialty.
- **color_group** has the weakest association of the four. As discussed in Section II.11, this finding (combined with its high missingness and limited information content) justified its removal from the feature set before modeling.

Overall, the chi-square ranking confirms that **region_group**, **variety_group**, and **processing_group** carry meaningful signal for predicting quality tier and should be retained as features in the logistic regression model in Section IV.

IV. Model Development

Based on the EDA findings, logistic regression was selected as the modelling algorithm. It is interpretable, appropriate for multi-class classification, and well-suited to the relatively small, mixed-type feature set produced by preprocessing. Four models were built to allow structured comparisons.

IV.1. Data Splitting

After preprocessing, the dataset of 1,251 rows was split into 80% training set and 20% testing set. A stratified sampling approach (**stratify = y**) was applied to maintain the same class distribution across both sets. Additionally, a fixed random seed (**random_state = 42**) was used to ensure that the results can be consistently reproduced.

Split	Total Records	Commercial	Premium	Specialty
Training (80%)	1,000	139 (13.9%)	780 (78.0%)	81 (8.1%)
Test (20%)	251	35 (13.9%)	196 (78.1%)	20 (8.0%)

IV.2. Model Selection & Training

Four models were selected to predict the quality of the coffee:

- **Dummy Classifier:** used as a baseline model by always predicting only “Premium” class. The classifier achieved an accuracy of 78.1% and a macro-F1 score of 0.29. Even though it got a very good result in the Premium class, it completely fails due to the “Commercial” and “Specialty” class, resulting in a very low macro-F1 score.
- **Model A: Logistic Regression using Production Features**
 1. Imbalanced Model: The model used default settings without class weighting. As a result, it tended to favor the dominant Premium class.
 2. Balanced Model: The model applied class weights, giving more balance to minority classes such as “Commercial” and “Specialty”, reducing bias toward the majority class.

Due to the strong class imbalance during exploratory data analysis, the balanced version was selected as the main Model A. It performed better in identifying Commercial and Specialty classes, whereas the unbalanced version behaved similarly to the baseline dummy classifier.
- **Model B: Logistic Regression using Flavor Score** The model was trained by using only one feature: **Flavor Score**. This model was expected to perform strongly due to the high correlation between **Flavor** and **Total Cup Points** ($r \approx 0.83$). It serves as a benchmark to evaluate how well a single key sensory attribute can predict coffee quality compared to a full set of production-related features.

V. Results and Evaluation

V.1. Performance Metrics

All models were evaluated using standard classification metrics, including **Accuracy**, **Macro-F1 score**, **Precision** and **Recall**. Accuracy measures the overall proportion of correct predictions, while Precision and Recall provide insight into how well each class is identified. However, due to the imbalance in the dataset across coffee quality classes, **Macro-F1 is the primary evaluation metric**, as it assigns equal weight to all classes and prevents the majority class (Premium) from dominating the overall performance evaluation.

Model	Accuracy	Macro F1	Precision	Recall
Dummy Classifier	0.781	0.292	0.260	0.333
Model A – Unbalanced	0.785	0.409	0.615	0.397
Model A – Balanced	0.526	0.463	0.475	0.644
Model B – Flavor Only	0.912	0.832	0.909	0.788

V.2. Model Comparison

- **Dummy Classifier:** The model achieves relatively high accuracy (78.1%) but a very low Macro-F1 score (0.292), indicating that it mainly predicts the majority class (Premium) while failing to correctly identify Commercial and Specialty. This confirms that accuracy alone is misleading in imbalanced classification tasks.
- **Model A – Unbalanced:** This model performs slightly better than the dummy baseline, with 78.5% accuracy and a Macro-F1 score of 0.409. While precision improves to 0.615, recall remains 0.397, showing that the model still favors the dominant Premium class and struggles to correctly identify minority classes. This suggests that production-related features alone provide limited predictive power when class imbalance is not addressed.
- **Model A – Balanced:** After applying class weighting, accuracy decreases to 52.6%, while the Macro-F1 score improves to 0.463, indicating more balanced performance across all classes. Recall also increases significantly to 0.644, showing improved identification

of *Commercial* and *Specialty* samples. This demonstrates that class weighting enhances minority class detection, even at the expense of overall accuracy.

- **Model B – Flavors only:** This model achieves the best overall performance, with 91.2% accuracy and a Macro-F1 score of 0.832. It also shows strong precision (0.909) and recall (0.788), indicating consistent performance across all classes. This demonstrates that the **Flavor** feature alone is highly predictive of coffee quality and outperforms models using multiple production-related features.

V.3. Confusion Matrix Analysis

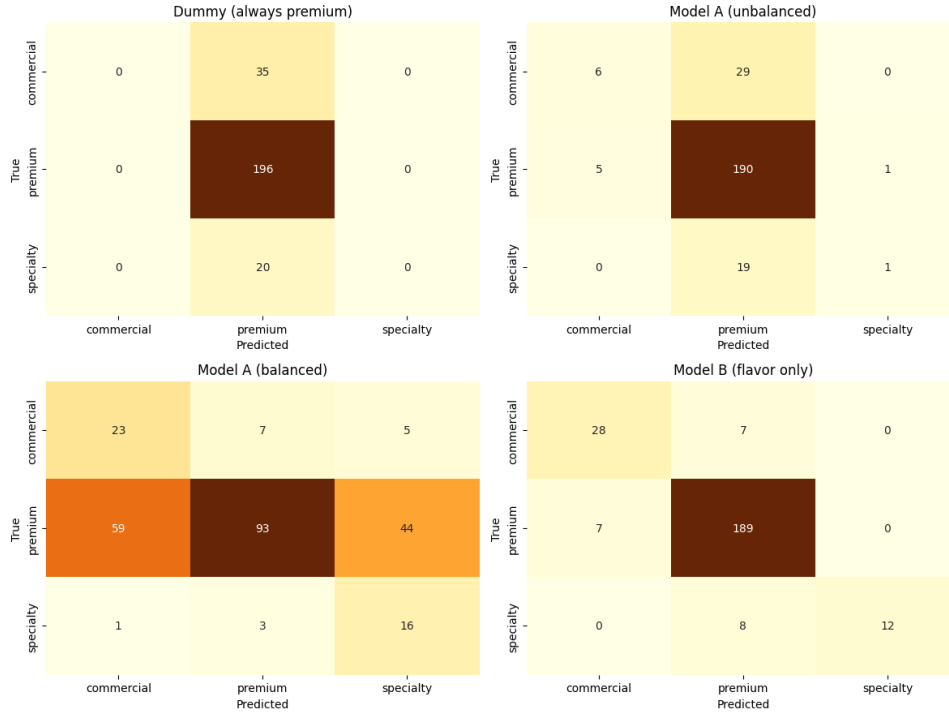


Figure 8: Confusion Matrices

The confusion matrices across all four models show that:

- **Dummy:** predicts every sample as “premium”, so it fails to identify “Commercial” and “Specialty” class.
- **Model A – Unbalanced:** correctly identifies most “Premium” samples, but misclassifies nearly all “Specialty” and “commercial” samples.
- **Model A – Balanced:** is able to identify more “Specialty” coffee, although it is still struggling to identify “Commercial” due to its small class.
- **Model B:** mostly correct predictions and slight confusion between “Premium” and “Specialty”.

V.4. Feature Importance

The coefficients from the balanced logistic regression model for the “Specialty” class were analyzed to identify the production features that most influence Specialty-grade classification. Positive coefficients indicate features that increase the likelihood of being classified as Specialty coffee.

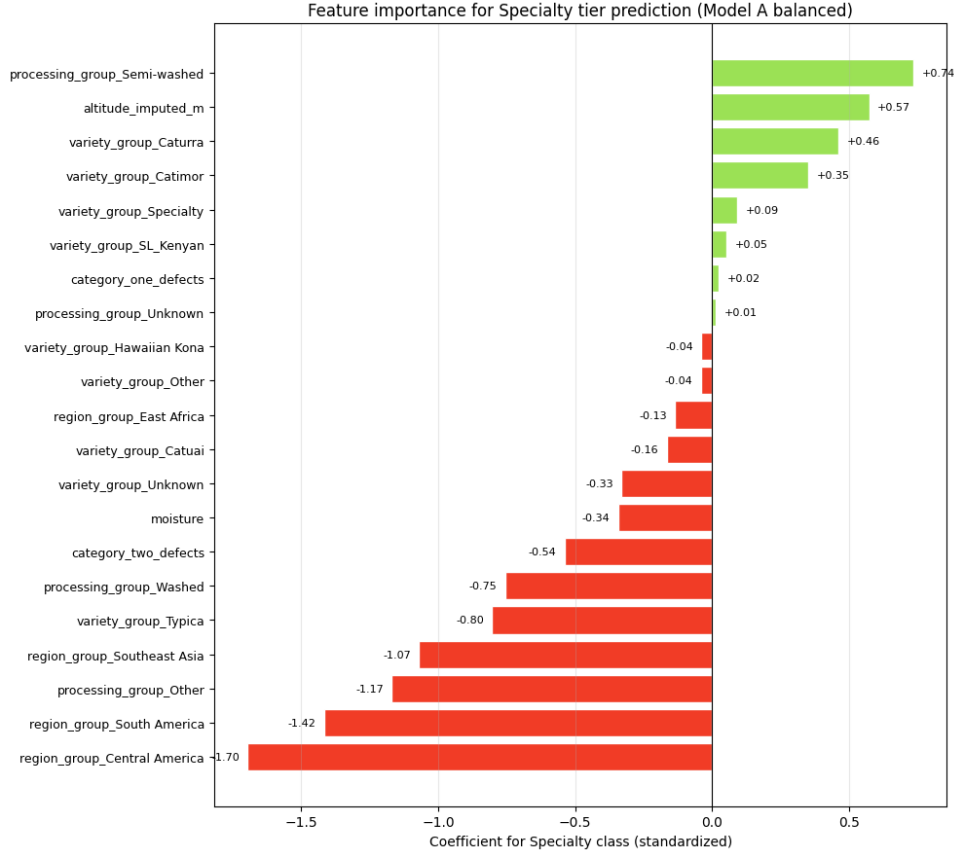


Figure 9: Feature Importance for Model A Balanced

The strongest positive predictors were:

- **processing_group_Semi-washed** (+0.74): semi-washed processing is the strongest positive predictor of Specialty classification, supporting the observation that minority processing methods tend to produce higher-quality coffee in this dataset.
- **altitude_imputed_m** (+0.57): higher altitude increases the probability of Specialty classification, consistent with the boxplots in Section III.4 and with established industry knowledge that altitude slows bean development and improves flavor.
- **variety_group_Caturra** (+0.46) and **variety_group_Catimor** (+0.35): both are positively associated with Specialty classification relative to the Bourbon reference category. While Catimor showed lower specialty rates in the EDA, its positive coefficient here reflects the model controlling for region and processing.

It is worth noting that **region_group_East Africa** and **variety_group_SL_Kenyan**, which appeared visually dominant in the bivariate plots (Section III.6), do not show strong positive coefficients in the model. This is due to multicollinearity: most SL Kenyan samples come from East African countries, so the model splits the predictive signal between the two related dummy variables. Additionally, the Caribbean/US reference region also has a high specialty rate (driven by Hawaiian Kona), which makes the East Africa coefficient look smaller than the visual evidence would suggest.

On the other hand, the strongest negative predictors were **region_group_Central America** (−1.70) and **region_group_South America** (−1.42), indicating that coffees from these regions are much less likely to be classified as Specialty compared to the Caribbean/US reference. **processing_group_Other** (−1.17) and **variety_group_Typica** (−0.80) also showed strong negative effects. The defect features **category_two_defects** and **category_one_defects** also reduce the likelihood of Specialty classification, which is consistent with industry expectations that defects

are direct quality penalties.

VI. Conclusion & Business Implications

This project shows that production-related information such as origin, variety, processing method, altitude, and defect counts can help predict coffee quality levels. The logistic regression model with balanced classes (Model A) performs much better than a simple baseline model, proving that it is possible to estimate coffee quality.

However, the gap between the production-feature model and the single sensory-score reference (Model B) is large. The best production model (Model A) achieves a Macro-F1 of approximately 0.46, compared to approximately 0.83 for the flavor-score model (Model B). This shows that cupping data provides much stronger predictive value than any mix of origin and production-related features included in this dataset.

For Business

- For buyers and importers: coffee from East Africa, especially when washed or naturally processed and with low defects, is often a strong sign of high quality. This allows buyers to focus on the most promising samples and reduce cupping time and costs (Iburu Coffee, 2021; Ebru Coffee Co., 2025).
- For farmers and cooperatives: lowering defect levels has the greatest effect on improving coffee grade. Better sorting and post-harvest processing tools can greatly increase the chance of reaching higher quality categories (SCA, 2005; World Economic Forum, 2020).
- For quality graders and certifiers: A simple screening model using basic origin and production data can reduce the need for full cupping, while still maintaining reliable classification results (Perfect Daily Grind, 2024).

Limitations:

- The dataset is small and has very few “Commercial” coffee samples, which can make the model less accurate.
- The model assumes simple relationships, so more advanced models might work better.
- Some values (like `altitude`) were estimated using country averages, which may reduce accuracy.

Future Works:

- Adding a small amount of sensory data could improve predictions a lot.
- Create a system that gives a real-time probability score for coffee quality to help business decisions.

VII. References

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Libraries:

- pandas (The pandas development team, 2020)
- numpy (Harris et al., 2020)
- matplotlib / seaborn (Hunter, 2007; Waskom, 2021)
- scipy.stats – chi-square tests (Virtanen et al., 2020)
- scikit-learn (Pedregosa et al., 2011)

VIII. Appendix

- **Link to GitHub Repository:** [Link](#)
- **Data Source:** [Link](#)